

Getting Started with GNU Radio

This article, which is targeted at scholars working in the area of digital signal processing and communication systems, gives an introduction to GNU Radio. It also covers the installation and a few uses of GNU Radio.



You must have listened to radio broadcasts from various radio stations. These stations send out radio waves, which are characterised by parameters like amplitude, frequency, phase, period, wavelength, etc, and need some kind of medium to travel through. Radio waves possess the maximum wavelength in the electromagnetic spectrum. Also, AM radio waves are longer in wavelength than television or FM radio waves.

Software Defined Radio

The term Software Defined Radio (SDR) was coined by Joseph Mitola in 1991. According to J. Bard and V.J. Kovarik (Reference 2), "SDR can be configured through the use of a personal computer equipped with a sound card, analog-to-digital converter, and some RF front end." In short, a software radio is a communication device created partially or entirely by using software.

Software Defined Radio brings together radio functionalities and signal processing capabilities, which helps to achieve reconfigurability and portability as a replacement to conventional hardware based radio. This enables researchers and designers to make dynamic changes frequently in software programs. Software radios can be

easily upgraded to new technologies by performing different software update downloads. They replace rigid hardware with flexible software based solutions.

Figure 3 depicts the tiered approach to transformation, from the conventional hardware based radio to the ultimate software based radio. Tier 0 (hardware based radio) is implemented using hardware components only, and does not offer any kind of portability. Tier 1 is configured using multiple hardware transceivers that support several protocols and software controllers, which are used to activate specific transceivers. A typical example of such type of radio is a dual SIM mobile phone, which implements both GSM and CDMA together. The second tier makes use of software to control various modulation techniques. GNU Radio and Universal Software Radio Peripheral (USRP) fall under this category. The next tier includes all the features of the previous tier, while offering analogue conversions to take place at the antenna mainly. The final tier is intended to have the intelligence to recognise traffic information and behave accordingly to optimise resources.

Many variants of SDR are available in the market. Some of them are commercial and others are open source. Cognitive Radio is an intelligent variant of SDR, which possesses